DS	**Session**	Arrays	**Question Information**	Level 1 Challenge 26

Problem Description:
Good news! Suresh get to go to America on a class trip! Bad news, he don't know how to use the Dollar which is the name of the American cash system. America uses coins for cash a lot more than the Kuwait does. Dollar comes in coins for values of: 1, 2, 10, 50, 100, & 500 To practice your Dollar skills, suresh have selected random items from Amazon.co.us and put them into a list along with their prices in Dollar. Suresh now want to create a program to check suresh Dollar math.

Suresh goal is to maximize your buying power to buy AS MANY items as you can with your available Dollar.

Input Format:
File listing 2 to 6 items in the format of:

ITEM DDDDD
ITEM = the name of the item you want to buy
DDDDD = the price of the item (in Dollar)

Output Format:
Print the output in a separate lines contains, List the items suresh can afford to buy. Each item on its own line. Suresh goal is to buy as many items as possible. If suresh can only afford the one expensive item, or 2 less expensive items on a list, but not all three, then list the less expensive items as affordable. If suresh cannot afford anything in the list, output "I need more Dollar!" after the items. The final line you output should be the remaining Dollar he will have left over after make purchases.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6000 3
Phone-case 1486
Candybar 863
Sunglasses 5529
```

EXPECTED OUTPUT

```
I can afford Phone-case
I can afford Candybar
I can't afford Sunglasses
3651
```

Test Case 2

INPUT (STDIN)

```
2100 3
Camera 69555
TV 76439
iPhone 90000
```

EXPECTED OUTPUT

```
I can't afford Camera
I can't afford TV
I can't afford iPhone
I need more Dollar!
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
char name[MAX][LEN];
```

Test Case 2

KEYWORD

```
int price[MAX]
```

Test Case 3

KEYWORD

```
afford[MAX]
```

Test Case 4

KEYWORD

```
for(i=0;i<items;i++)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

11

Test Case 2

TOKEN COUNT

250

Test Case 3

NLOC

33

Code

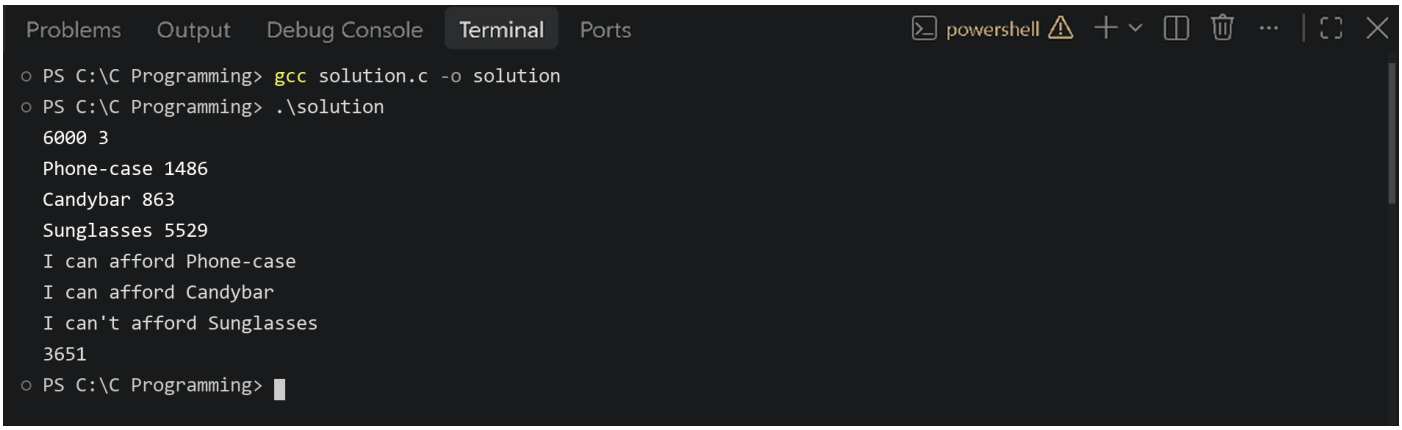
solution.c

```c
#include <stdio.h>
#include <string.h>

#define MAX 100
#define LEN 50

int main() {
    int budget, items;
    if (scanf("%d %d", &budget, &items) != 2) return 0;
    char name[MAX][LEN];
    int price[MAX];
    int afford[MAX];
    int i, j;
    for (i = 0; i < items; i++) {
        scanf("%s %d", name[i], &price[i]);
        afford[i] = 0;
    }
    int order[MAX];
    for (i = 0; i < items; i++) order[i] = i;
    for (i = 0; i < items - 1; i++) {
        for (j = i + 1; j < items; j++) {
            if (price[order[i]] > price[order[j]]) {
                int t = order[i]; order[i] = order[j]; order[j] = t;
            }
        }
    }
    int rem = budget, bought = 0;
    for (i = 0; i < items; i++) {
        int idx = order[i];
        if (price[idx] <= rem) {
            rem -= price[idx];
            afford[idx] = 1;
            bought++;
        }
    }
    for (i = 0; i < items; i++) {
        if (afford[i]) printf("I can afford %s\n", name[i]);
        else printf("I can't afford %s\n", name[i]);
    }
    if (bought == 0) printf("I need more Dollar!\n");
    else printf("%d\n", rem);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
6000 3
Phone-case 1486
Candybar 863
Sunglasses 5529
I can afford Phone-case
I can afford Candybar
I can't afford Sunglasses
3651
PS C:\C Programming>